Residual-noise calibration in off-grid OFDM: Leakage bias, temporal correlation, and cyclic-prefix controls

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Abstract

Pilot-residual noise calibration couples OFDM receiver decisions to the assumed channel subspace: excluded signal energy is counted as noise. We quantify when this leakage dominates calibration error and test the reconstruction gain available from changing the calibration source. Gaussian quadratic-form analysis separates squared leakage bias from finite-window variance, identifies their crossover for independent blocks, and quantifies the effect of temporally correlated gains. Delay-offset and subspace sweeps distinguish the benefit of suppressing leakage from the cost of losing residual noise dimensions. With 16 pilots and a 12-dimensional integer subspace, a separated fractional-delay profile at 20 dB gives a pooled noise estimate approximately 8.78 times the thermal variance. In an independent paired experiment with 900 groups per condition, an established cyclic-prefix difference statistic replaces residual calibration in the same continuous-delay receiver. At 20 dB SNR, normalized channel-estimation error improves by 2.98 dB and 4.25 dB for separated and closely spaced paths, respectively; paired 95% bootstrap intervals are [2.57, 3.34] and [3.97, 4.52] dB. A finite-burst waveform check verifies fractional-delay cancellation, and the experiment preserves the dependence between cyclic-prefix and FFT noise. Together, the analysis and controlled intervention distinguish calibration gaps that benefit from further averaging from those that call for a different noise source or channel representation.

Keywords: OFDM, noise variance estimation, fractional delay, channel estimation, cyclic prefix

1. Introduction

Noise calibration affects both the strength of channel denoising and the choice of a channel model in pilot-assisted orthogonal frequency-division multiplexing (OFDM). When no independent noise measurement is available, a receiver can estimate the noise variance from the portion of its pilot observations left unexplained by an assumed channel subspace. This economical approach couples calibration to the channel representation because unexplained signal components enter the same residual energy as receiver noise. Increasing the number of calibration blocks can therefore make a noise estimate more repeatable without making it more accurate.

This distinction matters when a short integer-delay representation is used for a channel with fractional path delays. The residual then contains channel energy outside the selected subspace. Increasing the subspace dimension, however, leaves fewer residual degrees of freedom for estimating noise. These effects create a trade-off among channel approximation, calibration variance, and observation cost. They also complicate the interpretation of downstream performance. A change in

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channel-estimation error may reflect a better representation, a different noise estimate, or a change in model order.

Established methods already address continuous channel parameters and unknown statistics. Parametric OFDM receivers avoid an integer-delay restriction [1]; variational line-spectrum estimation represents continuous-parameter uncertainty and learns noise statistics [2]. Newtonized pursuit and its multiple-snapshot extension provide further ways to refine continuous parameters [3, 4]. These methods motivate a specific diagnostic question: how much residual calibration error can additional observations reduce, and how much requires a change in the signal model?

We quantify when residual calibration becomes dominated by channel leakage and how temporal correlation changes the useful averaging window. The projected channel covariance separates persistent bias, receiver-noise fluctuations, and correlated leakage fluctuations. Their relative sizes give a model-conditioned crossover between sampling-dominated and bias-dominated calibration. Controlled OFDM comparisons then connect this distinction to channel reconstruction: changing the channel representation and changing the noise source address different parts of the error. A paired experiment substitutes cyclic-prefix (CP) calibration within the same continuous-delay receiver, using the existing waveform and preserving its noise dependence. The resulting diagnostic identifies when additional averaging is useful and when signal cancellation or a different channel representation is required.

2. Related work

2.1. Noise information and channel denoising

Risk-based denoising offers an established alternative to assuming a known channel covariance. Upadhya et al. use Stein's unbiased risk estimate to choose linear and nonlinear OFDM channel estimators [5]. In their accessible author preprint, the simulation setup estimates noise from 500 blank OFDM carriers. This differs from estimating noise from a small residual subspace of the pilots also used for channel estimation. Comparing these settings requires distinguishing the denoising rule from the amount and source of its noise information.

The CP provides noise information from samples already present in the waveform. Cui and Tellambura estimate the power delay profile and noise variance from the repeated signal structure; their Eq. (15) gives a direct CP–tail difference statistic [6]. Their simplified channel model rounds delays to integers and assumes synchronization and blockwise channel invariance. Here that established statistic serves as a calibration-source control: a conservative delay bound selects a clean repetition interval, a finite-burst waveform check verifies fractional-delay cancellation, and paired receiver comparisons retain the dependence between CP and FFT noise. This separates the source of noise information from the denoising rule that uses it.

2.2. Continuous parameters and multiple observations

Continuous-delay modeling has a direct counterpart in line-spectrum estimation. Newtonized orthogonal matching pursuit (NOMP) alternates detection with refinement of sinusoidal parameters and uses a noise-dependent stopping criterion [3]. Its multiple-snapshot extension, MNOMP, uses observations with shared frequencies and snapshot-dependent amplitudes [4]. For OFDM, Gao et al. exploit common delay structure and spatial and temporal correlations [7]. These approaches use shared channel structure across observations, whereas residual energy pooling shares a scalar noise statistic. The distinction matters when comparing observation budgets and attributing gains to calibration.

Unknown noise and continuous parameters can also be inferred jointly. VALSE maintains approximate frequency distributions, selects a model order, and updates the noise and gain statistics

Table 1: Closest prior approaches and the calibration question examined here. The rows distinguish information use, rather than numerical performance.

Approach	Established mechanism	Relation to this study
Risk-based denoising [5]	SURE selects channel denoisers; the author preprint uses 500 blank carriers for noise estimation.	Separates the denoiser from the quantity and source of noise observations.
CP calibration [6]	Direct CP–tail differences cancel repeated signal samples.	Supplies the established statistic for a paired calibration-source substitution with fractional delays.
NOMP/MNOMP and common-delay OFDM [3, 4, 7]	Continuous parameters and shared geometry improve channel fitting.	Distinguishes sharing channel structure from pooling residual energies.
VALSE and iterative OFDM receivers [2, 1, 10]	Joint inference learns continuous parameters, model order, and noise; iterative receivers also use decoder information.	Motivates the pilot-domain joint-inference check and fixes the information boundary of the receiver controls.

[2]. Hansen et al. incorporate parametric channel estimation into an iterative OFDM receiver that exchanges soft information with a decoder and represents dense channels through a virtual channel impulse response [1]. These methods establish joint continuous-parameter and noise inference; the calibration question here concerns the consequences of retaining a fixed residual subspace.

Subsequent variational formulations extend uncertainty-aware inference to multidimensional spectra [8] and noise variances that vary across sensors or snapshots [9]. For underwater acoustic OFDM, Wan et al. jointly estimate frequency offset, continuous-delay channels, and noise statistics while exchanging information with a decoder; their initialization also uses pilot-only VALSE [10]. These developments reinforce the distinction between learning nuisance parameters within a channel model and examining the consequences of fixing an imperfect calibration subspace.

Table 1 organizes these precedents by the information they use. Our contribution is the quantitative connection between a fixed subspace's leakage, the averaging window, and the reconstruction gain obtained by changing the calibration source. Standard Gaussian quadratic-form identities provide the analytical basis. A specified orthogonal matching pursuit (OMP)/variable-projection receiver provides the controlled intervention, and a separate paper-equation VALSE implementation examines joint inference in the pilot domain.

3. System model

3.1. OFDM observations and delay convention

Consider a single-antenna OFDM link with an even number N of subcarriers. Let $q_k = k$ for $0 \le k < N/2$ and $q_k = k - N$ otherwise, and set $f_k = q_k/N$ in cycles per base-rate sample. The frequency response in block t is

$$H_{t,k} = \sum_{j=1}^{J} g_{t,j} \exp(-\mathrm{i}2\pi f_k \tau_j), \qquad k = 0, \ldots, N-1, \tag{1}$$

where τ_j is measured in sample periods and delays remain fixed over a calibration window. The path gains are zero-mean proper complex Gaussian with covariance $\mathbf{P} = \mathrm{diag}(p_1, \ldots, p_J)$ and $\sum_j p_j = 1$. This normalization applies to the ensemble; individual channel realizations are not rescaled.

There are M equally spaced pilots of value one at indices $k_m = mD$, where $D = N/M$. After division by the known pilots, the observations satisfy

$$\mathbf{y}_t = \mathbf{B}\mathbf{g}_t + \mathbf{n}_t, \quad B_{mj} = \exp(-\mathrm{i}2\pi f_{k_m}\tau_j), \quad \mathbf{n}_t \sim \mathcal{CN}(\mathbf{0}, v\mathbf{I}_M). \tag{2}$$

Noise is independent across blocks and of all gains. The scalar v is the variance per complex pilot observation and is constant within the window. Under the normalization above, the nominal subcarrier SNR is $1/v$, excluding CP and pilot overhead. True delays, powers, and v are used for data generation and explicitly identified oracle references; practical estimators do not receive them.

The numerical setting uses $N = 64$, $M = 16$, and a nominal integer-delay window of length $L = 12$. For fractional delays, replacing a negative frequency by its unsigned FFT index changes the continuous-delay phase, so the signed-frequency convention matters. Pilot delay atoms have a period of M samples in this setting, and the predefined search interval $[0, 12]$ lies within one such period. All generators and estimators use the same convention.

3.2. Waveform assumptions and calibration windows

For a useful symbol and its CP, the ideal transmitted waveform is

$$s_t(u) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X_{t,k} \exp(\mathrm{i}2\pi f_k u), \qquad -N_{\mathrm{CP}} \le u < N. \tag{3}$$

Adjacent CP-extended symbols occupy separate rectangular intervals. With ideal delayed paths, exact synchronization, constant gains during the received symbol, and $N_{\mathrm{CP}} \ge \max_j \tau_j$, CP removal and a unitary FFT give the multiplicative response in (1). To check finite-CP consistency for ideal paths, the waveform audit applies linear delays to a finite oversampled burst rather than a circular shift to each symbol separately. The rectangular symbol boundaries do not form a globally bandlimited analog waveform. Transmit/receive filtering, frequency offsets, Doppler, and radio-frequency impairments are outside this model.

The main calibration experiment holds path geometry and noise variance fixed while drawing independent gains for each block; a separate analysis allows stationary temporal correlation. Independent fading blocks are not taken to be consecutive OFDM symbols, because the observation duration needed to obtain them depends on channel coherence. Only the first block is the target of channel estimation and data detection. The other blocks supply scalar calibration statistics unless a method is explicitly defined to fit shared channel parameters.

4. Analytical characterization of residual calibration

4.1. Fixed subspaces and conditional moments

Let $\mathbf{A} \in \mathbb{C}^{M\times L}$ contain the pilot responses of integer delays $0, \dots, L-1$, with $L < M$. For the comb pilots, $\mathbf{A}^{\mathrm{H}}\mathbf{A} = M\mathbf{I}_L$. Define

$$\mathbf{Q} = \mathbf{I}_M - \frac{\mathbf{A}\mathbf{A}^{\mathrm{H}}}{M}, \quad d = M - L, \quad \mathbf{z}_t = \frac{\mathbf{A}^{\mathrm{H}}\mathbf{y}_t}{M}, \quad R_t = \mathbf{y}_t^{\mathrm{H}}\mathbf{Q}\mathbf{y}_t. \tag{4}$$

$\mathbf{Q}$ is a fixed Hermitian projector with rank d. For a general full-rank pilot basis, the projector is instead $\mathbf{I}_M - \mathbf{A}(\mathbf{A}^{\mathrm{H}}\mathbf{A})^{-1}\mathbf{A}^{\mathrm{H}}$; the residual identities below remain valid. Pooling T blocks gives

$$\widehat{v}_T = \frac{1}{Td} \sum_{t=1}^{T} R_t. \tag{5}$$

First condition on the noiseless observations $\boldsymbol{\mu}_t = \mathbf{B}\mathbf{g}_t$. For proper complex Gaussian noise, expanding R_t into its deterministic, linear-noise, and quadratic-noise terms yields

$$\mathbb{E}[R_t \mid \boldsymbol{\mu}_t] = vd + \|\mathbf{Q}\boldsymbol{\mu}_t\|^2, \tag{6}$$

$$\mathrm{Var}(R_t \mid \boldsymbol{\mu}_t) = v^2 d + 2v\|\mathbf{Q}\boldsymbol{\mu}_t\|^2. \tag{7}$$

The linear term has variance $2v\|\mathbf{Q}\boldsymbol{\mu}_t\|^2$; the quadratic term has variance $v^2 \operatorname{tr}(\mathbf{Q}^2) = v^2 d$. Their covariance is zero because centered Gaussian third moments vanish. Since receiver noise is independent across blocks,

$$\mathbb{E}[\widehat{v}_T \mid \{\boldsymbol{\mu}_t\}] = v + \frac{\sum_t \|\mathbf{Q}\boldsymbol{\mu}_t\|^2}{Td}, \tag{8}$$

$$\mathrm{Var}(\widehat{v}_T \mid \{\boldsymbol{\mu}_t\}) = \frac{Tv^2 d + 2v\sum_t \|\mathbf{Q}\boldsymbol{\mu}_t\|^2}{T^2 d^2}. \tag{9}$$

If the signal belongs to the assumed subspace, all projected means vanish. Then $\sum_t R_t/v \sim \mathrm{Gamma}(Td, 1)$, so $\mathbb{E}[\widehat{v}_T] = v$ and $\mathrm{Var}(\widehat{v}_T) = v^2/(Td)$. These statements remain true for correlated gains when the subspace is correct, as the residual contains receiver noise alone. Increasing L reduces d, so a wider delay window increases calibration variance even without mismatch.

4.2. Ensemble leakage bias and Gaussian variance

To average over channel realizations, define

$$\mathbf{C}_s = \mathbf{B}\mathbf{P}\mathbf{B}^{\mathrm{H}}, \qquad \mathbf{S} = \mathbf{Q}\mathbf{C}_s\mathbf{Q}, \qquad \mathbf{C} = \mathbf{S} + v\mathbf{Q}, \qquad b = \frac{\operatorname{tr}(\mathbf{S})}{d}. \tag{10}$$

Here $\mathbf{S}$ is the covariance of the signal remaining after projection; $b \geq 0$ is its residual energy per retained complex dimension. Since $\mathbf{Q}^2 = \mathbf{Q}$, $\operatorname{tr}(\mathbf{S}) = \operatorname{tr}(\mathbf{Q}\mathbf{C}_s)$. The projected observation is proper Gaussian with covariance $\mathbf{C}$, possibly singular in the ambient pilot space. Diagonalizing this covariance expresses its squared norm as a weighted sum of independent unit-mean exponential variables, giving

$$\mathbb{E}[R_t] = \operatorname{tr}(\mathbf{C}), \qquad \mathrm{Var}(R_t) = \operatorname{tr}(\mathbf{C}^2). \tag{11}$$

The mean identity requires the stated second moments, whereas the variance identity also uses proper Gaussian gains. For independent blocks,

$$\mathbb{E}[\widehat{v}_T] = v + b, \tag{12}$$

$$\mathrm{Var}(\widehat{v}_T) = \frac{\operatorname{tr}(\mathbf{S}^2) + 2v\operatorname{tr}(\mathbf{S}) + v^2 d}{Td^2}. \tag{13}$$

Pooling reduces variance around $v+b$ but leaves b unchanged. For fixed channel power and geometry, the relative bias b/v increases with SNR. These quantities separate thermal-noise estimation error from residual signal-model error and can be computed before Monte Carlo simulation.

4.3. Correlated calibration blocks

Consider stationary Gaussian gains with a real autoregressive coefficient $0 \leq \rho \leq 1$:

$$\mathbf{g}_{t+1} = \rho\mathbf{g}_t + \sqrt{1-\rho^2}\,\mathbf{w}_t, \quad \mathbf{g}_1, \mathbf{w}_t \sim \mathcal{CN}(\mathbf{0}, \mathbf{P}), \tag{14}$$

where the innovations are mutually independent and independent of receiver noise. Thus $\mathbb{E}[\mathbf{g}_t\mathbf{g}_u^{\mathrm{H}}] = \rho^{|t-u|}\mathbf{P}$. For $t \neq u$, the residual cross-covariance is $\rho^{|t-u|}\mathbf{S}$. The proper Gaussian fourth-moment identity gives

$$\mathrm{Cov}(R_t, R_u) = \rho^{2|t-u|}\operatorname{tr}(\mathbf{S}^2). \tag{15}$$

Pairing the two cross-block factors in the expansion of $\mathbb{E}[R_t R_u]$ produces the trace of the cross-covariance times its Hermitian transpose; the within-block pairing cancels the product of means. Summing all diagonal and off-diagonal terms yields

$$\mathrm{Var}(\widehat{v}_T) = \frac{T\,\mathrm{tr}(\mathbf{C}^2) + 2\,\mathrm{tr}(\mathbf{S}^2)\sum_{h=1}^{T-1}(T-h)\rho^{2h}}{T^2 d^2}. \tag{16}$$

The expectation remains (12). To show which part of the variance is affected by correlation, set $a_T = T + 2\sum_{h=1}^{T-1}(T-h)\rho^{2h}$. Then

$$\mathrm{Var}(\widehat{v}_T) = \frac{v^2 d + 2v\,\mathrm{tr}(\mathbf{S})}{Td^2} + \frac{a_T\,\mathrm{tr}(\mathbf{S}^2)}{T^2 d^2}. \tag{17}$$

The effective sample size T^2/a_T describes the channel-leakage term, not the entire noise estimate. For fixed $\rho < 1$, $a_T/T \to (1+\rho^2)/(1-\rho^2)$, and the variance tends to zero while the estimate concentrates around $v + b$.

When $\rho = 1$, all blocks share one gain realization, $a_T = T^2$, and the ensemble variance instead approaches $\mathrm{tr}(\mathbf{S}^2)/d^2$. Conditional on that realization, averaging still removes receiver-noise fluctuations, and the estimate converges to $v + \|\mathbf{Q}\mathbf{B}\mathbf{g}_1\|^2/d$. An ensemble variance floor therefore does not mean that receiver-noise averaging has stopped working. In all cases, the mean-square calibration error is

$$\mathbb{E}[(\widehat{v}_T - v)^2] = b^2 + \mathrm{Var}(\widehat{v}_T), \tag{18}$$

showing the persistent contribution of bias.

4.4. A crossover for the value of additional averaging

For independent blocks and $b > 0$, write

$$A = \frac{\mathrm{tr}(\mathbf{S}^2) + 2v\,\mathrm{tr}(\mathbf{S}) + v^2 d}{d^2}, \qquad T_\star = \frac{A}{b^2}. \tag{19}$$

Equation (18) then becomes $b^2 + A/T$. The fraction attributable to squared bias is $\phi_T = b^2/(b^2 + A/T) = T/(T + T_\star)$. For $T \geq T_\star$, leakage bias accounts for at least half the mean-square calibration error; further pooling reduces its variance contribution. When $b = 0$, the error follows $v^2/(Td)$ throughout. Thus $T_\star$ locates a change in the dominant error source, conditional on the channel covariance and noise level. It is a design diagnostic rather than an estimated stopping rule.

For correlated gains, the same comparison uses (17) at the chosen (T, ρ). Correlation changes the leakage-fluctuation term while the receiver-noise terms retain their $1/T$ scaling. At $\rho = 1$, both the squared bias and the random channel-specific offset persist. These expressions distinguish three responses to a calibration gap: average while sampling variance dominates, change the signal subspace to alter leakage, or use signal-cancelling observations to estimate the thermal variance directly.

4.5. Dependence and fitted-model boundaries

In the correctly specified model, the target-block channel statistic and its residual noise are orthogonal. Under mismatch,

$$\mathbb{E}[\mathbf{z}_t(\mathbf{Q}\mathbf{y}_t)^{\mathrm{H}}] = \frac{\mathbf{A}^{\mathrm{H}}\mathbf{C}_s\mathbf{Q}}{M} \tag{20}$$

need not vanish. Although their receiver-noise components are independent, their signal components can couple channel estimation to calibration. Moving from $T = 1$ to $T > 1$ changes both the

dispersion of the noise estimate and its dependence on the target block, so a downstream error difference cannot be attributed entirely to variance reduction.

The moment identities describe a projector fixed independently of the observed data. For fitted delays or support, the projector is data-dependent and its residual distribution includes the fitting and selection process. Counting each fitted real delay as an additional real parameter gives a local degrees-of-freedom approximation whose accuracy depends on order selection and path resolution. Jointly fitted models therefore need their own uncertainty treatment, as in VALSE [2].

4.6. A cyclic-prefix calibration control

The CP–tail difference estimator of Cui and Tellambura [6, Eq. (15)] cancels the repeated signal before measuring noise energy. We use this established construction with a conservative memory bound to isolate calibration-source effects. With $N_{\mathrm{CP}} = 16$ and the known delay bound $\max_j \tau_j \le 12$, the last $K_{\mathrm{CP}} = 4$ base-rate CP samples and their counterparts at the end of the useful symbol have equal noiseless received values. For received time samples $r_t[u]$, define

$$\widehat{v}_{\mathrm{CP},T} = \frac{1}{2TK_{\mathrm{CP}}} \sum_{t=1}^{T} \sum_{a=1}^{K_{\mathrm{CP}}} |r_t[-a] - r_t[N-a]|^2. \tag{21}$$

Independent base-rate receiver noise makes each difference $\mathcal{CN}(0, 2v)$, giving exact moments

$$\mathbb{E}[\widehat{v}_{\mathrm{CP},T}] = v, \qquad \mathrm{Var}(\widehat{v}_{\mathrm{CP},T}) = \frac{v^2}{TK_{\mathrm{CP}}}. \tag{22}$$

Using existing CP samples and the declared delay bound, this control adds no transmitted pilots. It requires synchronization, an adequate channel memory margin, and unchanged gains over each paired interval.

The paired noise differences must remain coupled to the pilot observations. If $\mathbf{F}$ is the unitary FFT matrix, $\mathcal{I}$ indexes the paired useful-symbol samples, and $\mathbf{e} = (\mathbf{n}_{\mathrm{CP}} - \mathbf{n}_{\mathcal{I}})/\sqrt{2}$, then

$$\mathbb{E}[\mathbf{n}_{\mathcal{P}} \mathbf{e}^{\mathrm{H}}] = -\frac{v}{\sqrt{2}} \mathbf{F}_{\mathcal{P},\mathcal{I}}, \tag{23}$$

where $\mathbf{n}_{\mathcal{P}}$ denotes the FFT noise on the pilots. A separate independent calibration sample with the same marginal distribution would discard this dependence. We therefore form CP differences and FFT observations from the same time-domain noise in the receiver comparison.

5. Experimental design

5.1. Channel profiles and information budgets

Table 2 lists five controlled model profiles. The two dense profiles test correctly specified integer subspaces with different power-delay spreads, and the sparse integer profile examines the effect of sparsity without fractional delays. The remaining profiles introduce separated fractional paths and a closely spaced pair. These controlled profiles isolate delay geometry, sparsity, and power spread rather than represent a standardized channel family. All channel and noise draws are independent between Monte Carlo groups. Within a group, 16 blocks share delays, powers, and noise variance. Their gains are independent except in the correlation audit.

Each target block contains 16 pilots at FFT indices $0, 4, \ldots, 60$ and 48 unit-energy QPSK data symbols, with a CP of 16 base-rate samples. The noise variance is $v = 10^{-s/10}$ for nominal SNR $s \in \{0, 10, 20\}$ dB. Residual calibration uses $T = 1, 4$, or 16 blocks with $d = 4$. A separate control

Table 2: Simulated power-delay profiles. Delays are in base-rate sample periods; all powers sum to one. The receiver is not given the profile identity or true path count.

Profile	Delays	Relative powers
Exp-2	$0, 1, \ldots, 11$	$p_j \propto e^{-j/2}$
Exp-8	$0, 1, \ldots, 11$	$p_j \propto e^{-j/8}$
Sparse integer	$0, 7, 11$	$0.5, 0.3, 0.2$
Separated fractional	$0, 7.35, 11.2$	$0.5, 0.3, 0.2$
Close fractional	$0, 3.1, 3.45, 10.7$	$0.4, 0.2, 0.2, 0.2$

Table 3: Information resources of the numerical controls. Every channel estimate uses the same target pilots within a paired comparison. Scalar pooling shares noise statistics across blocks; it does not jointly fit their channel geometry.

Control	Calibration data and sharing	Supplied model information	Reported metric
Residual	T blocks of 16 pilots; residual energies pooled	Fixed integer subspace; refined fit uses $[0, 12]$	Noise; full-channel NMSE/BER
Extended EB	16 pilot blocks; one residual dimension per block	Integer columns $-2, \ldots, 12$	Full-channel NMSE/BER
CP	Four CP–tail pairs per block, 16 blocks; energies pooled	Delay bound 12 and a clean repetition interval	Noise; full-channel NMSE/BER
Noise probes	64 additional signal-free complex samples	Stationary noise variance	Noise calibration
Known noise / oracle	True v; target pilots for channel estimation	Refined: $[0, 12]$; oracle: exact channel covariance	Full-channel NMSE/BER
VALSE	One target pilot block; joint noise and channel inference	Circular frequency support; inferred order and noise	Noise; pilot-domain NMSE

uses 64 independent signal-free complex noise samples. These additional observations are counted separately from the pilot budget. The CP control uses four already transmitted repeated samples per block, together with the useful-symbol samples and a known delay upper bound of 12. The known-noise reference receives the true v; the oracle LMMSE reference also receives the exact physical channel covariance. The practical controls fit the channel using data from the target block alone.

5.2. Specified channel estimators

Let $\mathbf{F}_L$ denote the response of integer delays $0, \ldots, L-1$ on all subcarriers. The delay-domain least-squares estimate is $\widehat{\mathbf{H}} = \mathbf{F}_L \mathbf{z}_1$. The main inexpensive control uses positive-part empirical-Bayes (EB) shrinkage,

$$\widehat{h}_\ell = z_{1,\ell} \left(1 - \frac{\widehat{v}}{M |z_{1,\ell}|^2} \right)_+, \qquad \widehat{\mathbf{H}} = \mathbf{F}_L \widehat{\mathbf{h}}, \tag{24}$$

with zero output when $z_{1,\ell} = 0$. A cyclically extended control uses the 15 integer-delay columns $-2, \ldots, 12$, leaving one residual complex dimension per block. It uses the same shrinkage rule with its own 16-block residual statistic, so the extension changes both representation and calibration. Uniform Gaussian shrinkage and a fixed bank of exponential power-delay priors are retained in the numerical archive as auxiliary controls. The main comparisons use the EB rule defined above.

The sparse continuous-delay control is a generic OMP/variable-projection implementation. It starts at order zero and selects each new atom from the grid $\{0, 0.25, \ldots, 12\}$ by maximum residual

correlation. It then solves for complex gains by least squares and jointly refines all selected delays in $[0, 12]$. Variable projection eliminates the gains. The bounded dogbox optimizer uses an analytic real-delay Jacobian and allows at most 60 objective evaluations per candidate, with tolerances 10^{-9} for function/step changes and 10^{-10} for the gradient. Any refinement that increases the residual norm is rejected. All visited orders, up to six paths, are compared using

$$\text{score}(k) = \frac{2\,\text{RSS}_k}{\widehat{v}} + d_k \log(2M), \qquad d_k = \begin{cases} 2k, & \text{fixed-grid gains,} \\ 3k, & \text{continuous delays and gains.} \end{cases} \tag{25}$$

The search can stop when the next order's penalty alone exceeds the current minimum score. A scale-dependent numerical floor is used solely to avoid division by zero. Equation (25) is a BIC-type heuristic with a supplied noise estimate; its finite-sample behavior depends on discrete search multiplicity and path resolution. The grid control omits continuous refinement and uses $d_k = 2k$, so comparison with the refined control changes both the fit and the parameter penalty.

The oracle reference is

$$\widehat{\mathbf{H}}_{\text{or}} = \mathbf{C}_{H,\mathcal{P}}(\mathbf{C}_{\mathcal{P},\mathcal{P}} + v\mathbf{I}_M)^{-1}\mathbf{y}_1, \tag{26}$$

where $\mathbf{C}_H$ is generated from the true delays and powers. This reference quantifies the performance available with exact channel covariance and noise.

5.3. Replication, metrics, and numerical validation

The broad calibration sweep uses three seeds with 5000 groups per cell (15 000 total). The full reconstruction sweep uses three different seeds with 300 groups per cell (900 total). The correlation audit uses 20 000 groups for each $\rho \in \{0, 0.9, 0.99, 1\}$, with paired $T = 1, 4, 16$ statistics. These sweeps are exploratory. After inspecting them, we fixed two fractional profiles, SNRs 10 and 20 dB, all estimator settings, and three new seeds for a separate CP-control protocol. We then generated 300 groups per seed and cell, without using the new observations for estimator tuning. Within each experiment, all methods receive identical target channels, pilot noise, and data symbols.

Full-channel normalized mean-square error is computed from aggregate energies,

$$\text{NMSE} = \frac{\sum_{i=1}^{G} \|\widehat{\mathbf{H}}_i - \mathbf{H}_i\|^2}{\sum_{i=1}^{G} \|\mathbf{H}_i\|^2}. \tag{27}$$

Per-realization ratios are not averaged. Uncoded BER uses the signs of the real and imaginary parts of $Y_k \widehat{H}_k^*$ on the 48 data tones, giving 96 bits per target block. Bootstrap intervals are obtained by resampling whole groups with the same indices for both methods. Positive gain denotes $10\log_{10}(\text{MSE}_{\text{reference}}/\text{MSE}_{\text{candidate}})$. The exploratory reconstruction intervals use 400 resamples; the independent CP contrasts use 2000. These are pointwise percentile intervals without multiplicity adjustment; bits within one fading realization are not treated as independent bootstrap observations. We summarize noise calibration by the mean ratio $\mathbb{E}[\widehat{v}]/v$ and the relative standard deviation $\sqrt{\text{Var}(\widehat{v})}/v$, estimated across groups. The relative standard deviation uses the true thermal variance as its denominator rather than the potentially biased mean estimate.

The CP experiment generates independent frequency-domain Gaussian receiver noise, transforms it by a unitary IFFT to obtain the useful-symbol tail, and combines that tail with independent CP noise. The same frequency-domain noise is used on pilots and data. This construction is equivalent to generating white base-rate time noise and preserves the dependence in (23). An independent check of signal cancellation uses linear path shifts of a finite CP-extended waveform at oversampling factor 20. All delays in Table 2 lie exactly on this oversampling grid. In this audit, the fractional-delay and integer-delay FFT-response discrepancies are below 9×10^{-15}. Shortening

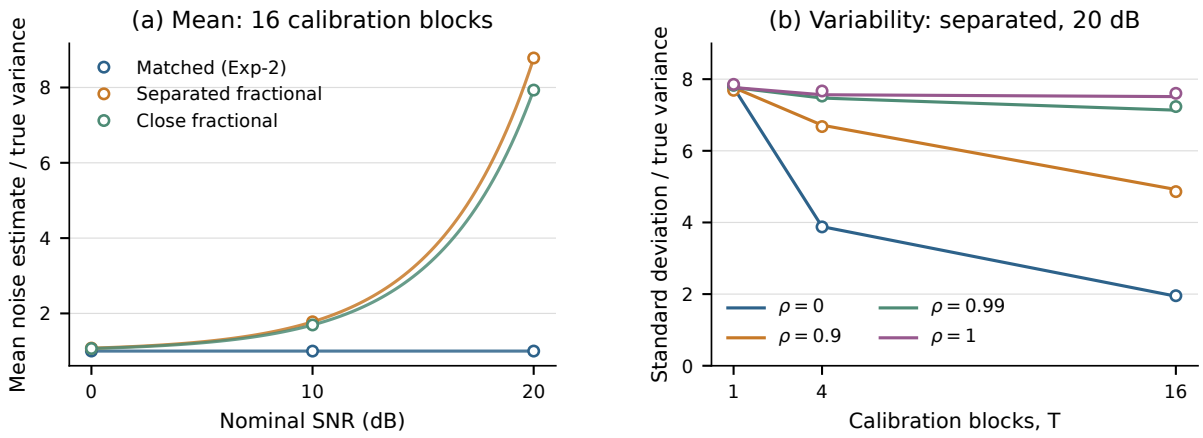


Figure 1: Residual-noise calibration. (a) Empirical 16-block means (markers, 15 000 groups per condition) and analytical expectations (curves). (b) Empirical relative standard deviations (markers, 20 000 groups per correlation condition) and Gaussian predictions (lines joining the evaluated block counts), for the separated fractional profile at 20 dB. Correlation acts on the leaked channel energy; the mean remains biased at every block count.

the CP to eight samples produces a relative RMS discrepancy of 0.151 for the fractional example. The four CP signal pairs also cancel to numerical precision in every profile.

Experiments use Python 3.13.9, NumPy 2.1.3, and SciPy 1.17.1 on CPU, with one BLAS thread per process. No learned model or GPU training is used. Source hashes, seeds, per-group errors, and optimizer diagnostics are retained. Numerical tests check projection moments, variable-projection derivatives, noiseless recovery, the CP control, and the variational implementation described below. OpenAI Codex was used for debugging assistance.

6. Results

6.1. Averaging controls variance, not fractional-delay leakage

Figure 1 compares predicted and empirical calibration statistics. The separated fractional profile has a projected signal bias of $b = 0.0777473$, giving an expected noise ratio at 20 dB of $1 + b/0.01 = 8.7747$. Across 15 000 groups, the measured ratios for $T = 1, 4, 16$ are 8.754, 8.784, and 8.783, while their relative standard deviations fall from 7.673 to 3.897 and 1.927. The matched Exp-2 profile has a 16-block ratio of 1.0002 and relative standard deviation 0.1242, close to $1/\sqrt{64} = 0.125$. The additional 64 independent noise probes also give a ratio close to one. Finite noise-sample variance alone therefore does not explain the discrepancy under fractional delays.

The gain from pooling depends strongly on temporal correlation. In the separate correlation audit, the measured relative standard deviation at $T = 16$ is 1.955 for independent gains, 4.861 for $\rho = 0.9$, 7.235 for $\rho = 0.99$, and 7.605 for constant gains. The corresponding analytical values are 1.942, 4.920, 7.135, and 7.513. The observations agree with the separate receiver-noise and channel-leakage terms in (17). Thus the benefit of a 16-block window depends on its temporal spacing and gain correlation.

For the integer EB receiver, increasing residual pooling from one to 16 blocks in the separated fractional 20 dB case improves NMSE by 0.047 dB (pointwise 95% interval [0.029, 0.064] dB), while BER increases by 0.00067 (95% interval [0.00050, 0.00083]). A residual estimate can thus become sharply concentrated while a substantial reconstruction error floor remains.

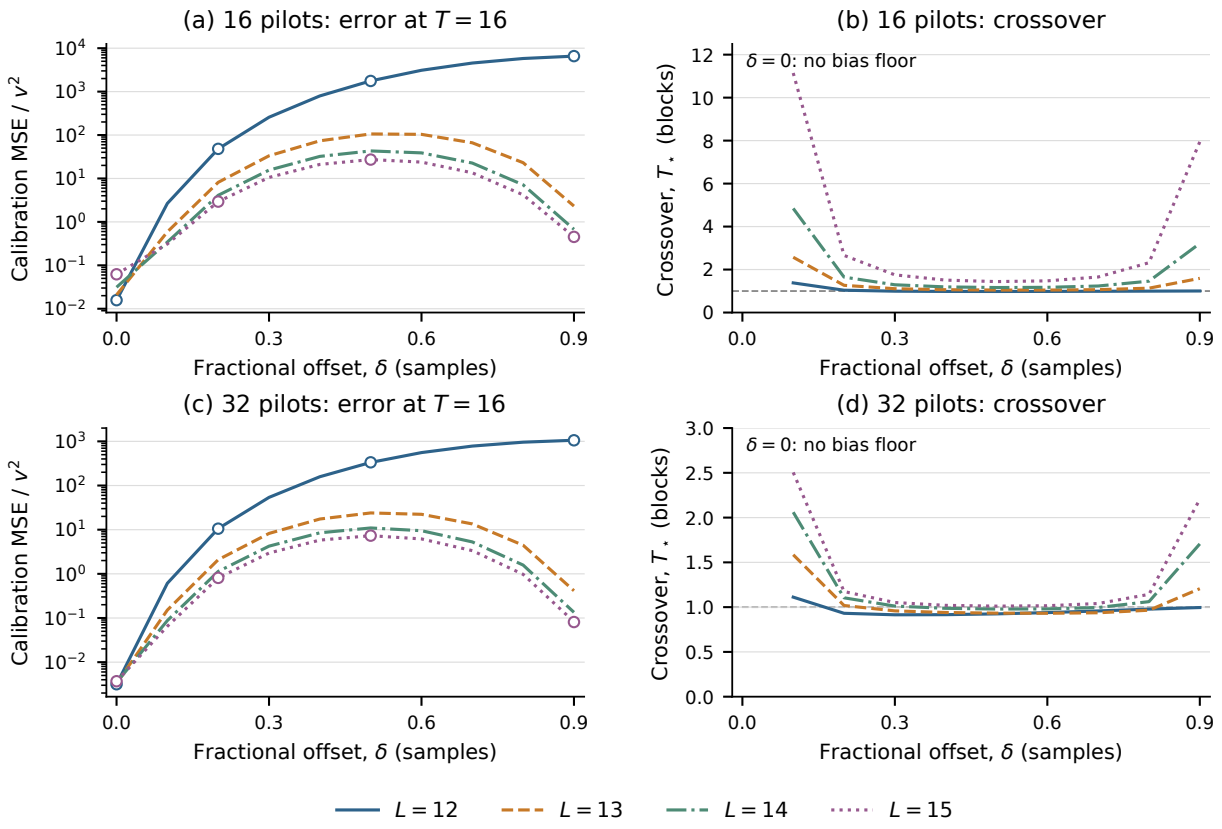


Figure 2: Geometry and window sensitivity at 20 dB with independent gains. Left: normalized calibration MSE at $T = 16$; curves give the analytical grid and open circles the Monte Carlo checks for $L = 12, 15$ (20 000 groups per condition). Right: the bias–variance crossover $T_\star$ for nonzero offsets. At $\delta = 0$, the channel lies in every tested subspace and there is no finite crossover. The dashed horizontal reference marks one block. Lines connect evaluated offsets; the full archive also contains 10 dB and $T = 1, 4$ results.

6.2. Delay offset, subspace width, and useful averaging

A prespecified sensitivity study varies the pilot count, delay offset, and integer-window length. It uses $N = 64$, $M \in \{16, 32\}$ equally spaced pilots, delays $(0, 7 + \delta, 11 + \delta)$, powers $(0.5, 0.3, 0.2)$, $\delta = 0, 0.1, \ldots, 0.9$, and $L = 12, 13, 14, 15$. The 160 combinations of geometry and SNR (10 and 20 dB) are evaluated analytically for $T = 1, 4, 16$. Independent Monte Carlo checks use 20 000 groups per condition at 20 dB for both pilot counts, $L = 12, 15$, and $\delta = 0, 0.2, 0.5, 0.9$. The complete grid and group estimates are archived.

Figure 2 shows how the crossover changes with the channel geometry. Here normalized calibration MSE means $\mathbb{E}[(\hat{v}_T - v)^2]/v^2$, distinct from the full-channel NMSE in Eq. (27). At $M = 16$, $\delta = 0.5$, and 20 dB, increasing L from 12 to 15 reduces b/v from 40.79 to 5.00 and normalized calibration MSE at $T = 16$ from 1765.78 to 27.24. The corresponding $T_\star$ values are 0.98 and 1.44 blocks: both estimates are bias-dominated well before 16 independent blocks. With 32 pilots, the same window change reduces normalized MSE from 333.46 to 7.31. Across the 48 Monte Carlo checks (16 conditions, three block counts), means and mean-square errors differ from their predictions by at most 2.91 and 2.79 Monte Carlo standard errors; sample variances differ by at most 3.02%.

Wider windows reduce both raw leakage and normalized bias at every fractional offset in this

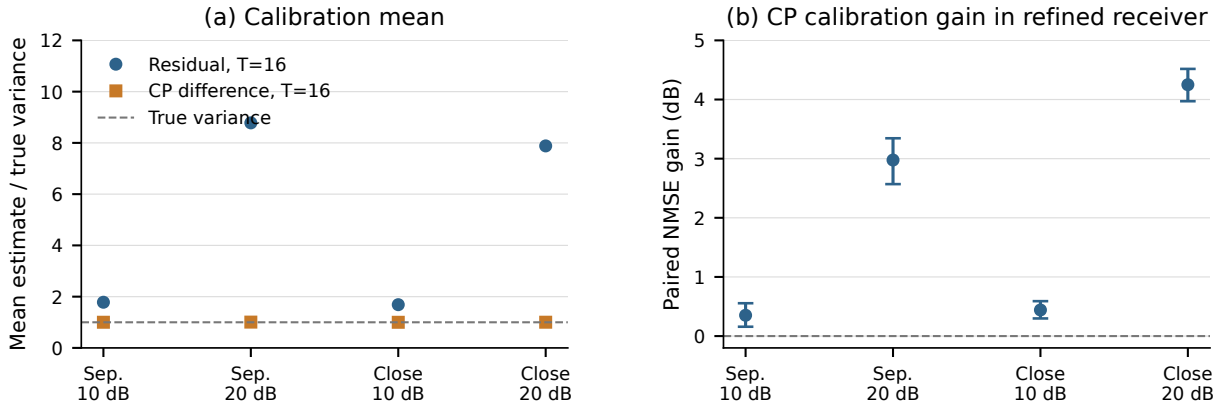


Figure 3: Independent CP-control experiment: 900 groups per cell from three new seeds. (a) Noise-calibration ratios. (b) NMSE gain obtained by replacing 16-block pilot-residual calibration with 16-block CP calibration in the same refined receiver. Error bars are pointwise 95% paired bootstrap intervals from 2000 resamples. CP and FFT noise retain their physical dependence.

grid, but the calibration MSE also depends on the remaining residual dimensions. At $M = 16$, $\delta = 0.1$, and 10 dB, changing L from 12 to 15 lowers b/v from 0.157 to 0.043 while increasing normalized MSE at $T = 16$ from 0.0465 to 0.0698. The intermediate choice $L = 13$ gives 0.0292, the smallest MSE among the four tested widths in this condition. For the matched integer case, the same change increases MSE from 1/64 to 1/16. Thus a wider window is beneficial when the leakage reduction compensates for the variance cost. The crossover and MSE quantify that trade-off for the specified channel family and noise level.

6.3. Confirmation with cyclic-prefix calibration

The paired CP experiment changes the received calibration samples while keeping the receiver, its search settings, and the target channel and pilot realization fixed. It uses the existing CP and its useful-symbol counterparts. Figure 3 shows all four prespecified conditions. CP noise ratios range from 0.9999 to 1.0069, with relative standard deviations 0.121–0.129, consistent with the predicted value 0.125. At 20 dB, the corresponding residual ratios are 8.781 and 7.882 for separated and close fractional paths.

For separated paths at 20 dB, the refined receiver changes from −21.76 dB with residual calibration to −24.74 dB with CP calibration; the paired gain is 2.98 dB [2.57, 3.34]. For close paths, it changes from −19.15 to −23.41 dB, a gain of 4.25 dB [3.97, 4.52]. BER changes from 0.01000 to 0.00662 and from 0.01206 to 0.00756, respectively. The remaining NMSE differences from the known-noise controls are 0.10 and 0.04 dB. At 10 dB, CP gains are smaller, 0.35 dB [0.16, 0.56] and 0.44 dB [0.30, 0.59], consistent with the smaller leakage-to-noise ratio.

The integer EB receiver gains 0.82 and 0.84 dB at 20 dB on the same observations. Its CP-calibrated NMSE remains −13.93 and −13.66 dB. The smaller gain leaves a substantial representation-dependent error. In the continuous receiver, the mean selected order rises from 2.73 to 3.11 for separated paths and from 2.99 to 3.86 for close paths when CP calibration replaces residual calibration. The changes are consistent with the residual estimate suppressing weaker components through the score in (25). These statistics describe the selected orders as receiver decisions rather than estimates of an exactly identified path count.

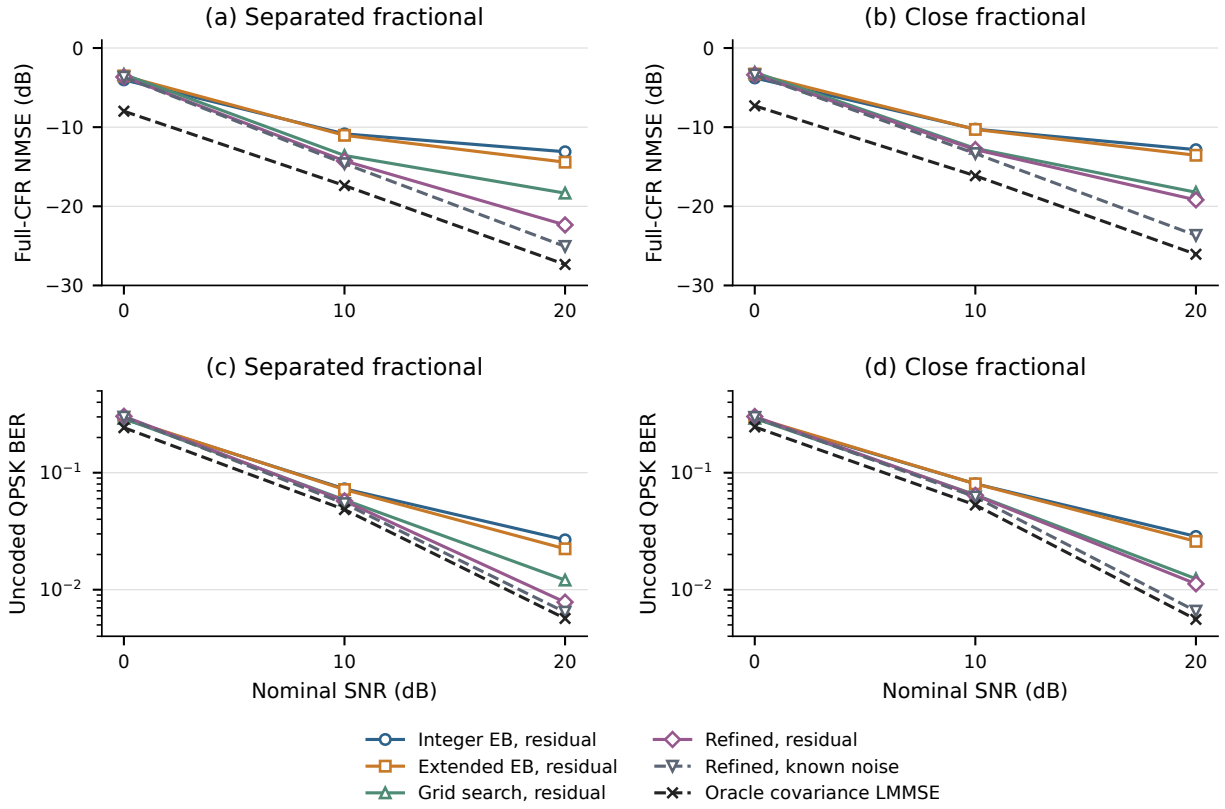


Figure 4: Exploratory full-channel reconstruction and uncoded detection with 900 paired groups per cell. Residual controls use 16 calibration blocks; the extended integer control computes its own one-dimensional residual. The grid and refined controls have different order penalties. Known noise and exact-covariance LMMSE are explicitly identified information-rich references.

The main calibration contrast is stable under an increased optimization budget: rerunning the flagged fits changes each primary 20 dB CP gain by less than 10^{-4} dB. The original setting selected a candidate at its evaluation limit in 121 of 10 800 refined fits; four such cases remained after the budget extensions. Appendix Appendix B gives the full diagnostic counts.

6.4. Representation and calibration have distinct effects

Figure 4 and Table 4 compare the controls for full-channel reconstruction. In the separated fractional 20 dB case, extending the integer window improves NMSE from −13.11 to −14.43 dB. Continuous-delay fitting with the original pooled residual statistic reaches −22.36 dB, a paired gain of 7.93 dB over the extended EB control (95% interval [7.54, 8.38] dB). Supplying the true noise variance to the same continuous receiver improves NMSE further to −25.07 dB. Changing the representation removes much of the error, but contaminated calibration still affects model-order selection.

The gap is larger for the closely spaced profile, where the refined residual control gives −19.19 dB and the known-noise version −23.67 dB. The dense profiles favor the integer subspace. The six-path refined control gives −15.55 and −9.94 dB for Exp-2 and Exp-8, whereas integer EB gives −21.16 and −21.13 dB. The path cap is below the 12 generating paths in these cases. The dense-profile comparison is therefore specific to this six-path fit and its selection and optimization settings.

Table 4: Full-channel NMSE (dB) at 20 dB nominal SNR; 900 paired groups per profile. Integer EB, extended EB, and grid/refined residual controls use 16-block calibration. The known-noise and oracle columns receive extra statistical information.

Profile	Integer EB	Extended EB	Grid residual	Refined residual	Refined known v	Oracle LMMSE
Exp-2	-21.16	-20.91	-13.65	-15.55	-15.55	-21.54
Exp-8	-21.13	-20.88	-9.09	-9.94	-9.94	-21.29
Sparse integer	-25.25	-24.66	-24.87	-25.28	-25.31	-27.57
Separated fractional	-13.11	-14.43	-18.34	-22.36	-25.07	-27.35
Close fractional	-12.84	-13.55	-18.23	-19.19	-23.67	-26.06

Table 5: Pilot-domain reconstruction and noise inference; 450 groups per cell. Columns 3–6 give NMSE in dB. VALSE learns noise from one block with circular frequency support; refined controls use the delay bound $[0, 12]$. The comparison concerns the observed pilot grid.

Profile	SNR (dB)	Integer EB	Refined residual	Refined known v	VALSE	VALSE $\mathbb{E}[\widehat{v}]/v$
Sparse integer	10	-14.86	-14.44	-14.55	-14.30	1.04
Sparse integer	20	-24.98	-24.86	-24.86	-23.86	1.13
Separated fractional	10	-10.95	-14.09	-14.57	-14.46	1.04
Separated fractional	20	-13.76	-22.13	-24.97	-23.92	1.17
Close fractional	10	-10.75	-12.76	-13.24	-12.97	1.12
Close fractional	20	-13.82	-18.92	-23.83	-19.85	1.94

Continuous-delay models can also represent dense channels through a richer virtual channel impulse response, as demonstrated by Hansen et al. [1].

6.5. Joint inference with a continuous probabilistic model

Joint continuous-parameter inference provides a second route to separating signal and noise. We evaluate a Python implementation of VALSE Heuristic 2 from the paper equations [2], retaining frequency and gain uncertainty in the noise update. VALSE learns from one target block over a circular frequency support, while the refined control uses the delay bound $[0, 12]$ and 16 calibration blocks for its residual statistic. The comparison therefore measures pilot-domain reconstruction and inferred noise under these different resources. Full-channel interpolation would additionally require a physical delay branch for the frequency posterior.

Three new seeds give 450 paired groups for each of three sparse profiles at 10 and 20 dB. The primary setting uses a 500-iteration limit and a relative reconstruction-change tolerance of 10^{-6}. All 2700 outputs are retained: 2688 met the stopping criterion, and 12 reached the iteration limit. Appendix Appendix A details the implementation and numerical checks. Replaying the 12 capped inputs with a 5000-iteration ceiling gives 11 further convergences. Across the six cells, pilot NMSE changes by at most 0.000624 dB and the mean inferred-noise ratio by 2.72×10^{-5}. The primary table retains the original 500-iteration outputs.

Table 5 shows noise inference with a continuous probabilistic model. At 20 dB, the mean noise ratios were 1.13, 1.17, and 1.94 for integer, separated fractional, and close fractional paths, respectively. The larger close-pair ratio accompanies a larger pilot reconstruction error, identifying a condition where the specified joint inference retains appreciable calibration error. For separated fractional paths, pilot NMSE was -23.92 dB, between the residual-calibrated and known-noise refined controls. These pilot-domain observations complement the full-channel CP intervention by showing how calibration also depends on the inferential model.

7. Discussion

Calibration-source substitution separates a recoverable noise-calibration gap from the error imposed by the channel representation. In the specified continuous-delay receiver, CP calibration recovers most of the performance available with known noise. Applying the same statistic to integer EB yields a smaller gain because its representation still excludes fractional-delay energy. This paired contrast links the projected leakage identified by the analysis to a consequential receiver decision: an inflated noise estimate changes the penalty for retaining weak components.

The bias–variance crossover makes the averaging trade-off explicit. Once b^2 exceeds the variance at the selected window, additional residual samples mainly sharpen an estimate centered above the thermal variance. Temporal correlation extends the window needed to suppress leakage fluctuations; with constant gains, the estimate stabilizes around a channel-specific offset. Changing the integer window alters both leaked energy and residual dimension, so its calibration effect is determined by their ratio. These quantities provide a common basis for comparing more observations, a different subspace, and a signal-cancelling noise source.

Information budgets determine how these alternatives can be used. The CP control uses additional received samples from the existing waveform and a known memory bound. Independent probes supply separate signal-free observations. Shared-delay methods additionally exploit channel structure across blocks [4, 7]. Accordingly, an observation window is characterized by its pilots, waiting duration, stationarity assumptions, and shared parameters. Within the present paired CP comparison, the channel data and receiver settings are held fixed while the calibration source changes; the resulting effect includes model-order selection through Eq. (25). The grid and refined controls also change parameter penalties and serve as model comparisons.

The scope is fixed-subspace calibration under the stated synchronized OFDM model and its explicitly specified receiver controls. The finite-burst check validates fractional delays for ideal paths; filtering requires the corresponding effective channel and noise covariance. CP cancellation requires a clean repeated interval, sufficient effective memory margin, and unchanged gains across each pair. The dense-channel comparison concerns the chosen six-path control and its selection and optimization settings; parametric receivers with richer representations can also model dense channels [1]. The separate VALSE experiment examines pilot-domain joint inference, with its own support and observation budget. These boundaries connect each numerical result to the calibration or representation choice that generated it.

8. Conclusion

Residual calibration has a quantifiable transition from sampling-dominated to leakage-dominated error. For independent blocks, its mean-square error is $b^2 + A/T$, so $T_\star = A/b^2$ for $b > 0$ marks equal contributions from squared bias and variance. Correlated gains change the leakage-fluctuation term and the useful averaging window. The geometry sweep demonstrates the associated subspace trade-off: reducing leakage can cost residual dimensions and increase finite-window error near a matched, low-SNR setting.

The paired receiver experiment establishes the consequence of changing the noise source. With target pilots and the continuous-delay receiver held fixed, the established CP statistic improves NMSE by 2.98 and 4.25 dB at 20 dB for the two fractional profiles and recovers most of their known-noise performance. Integer EB retains a larger representation error on the same observations. These results provide a model-conditioned basis for choosing between additional averaging, signal-cancelling calibration, and a more suitable channel representation.

Appendix A. Variational implementation and numerical sensitivity

The VALSE implementation follows Heuristic 2 and the paper equations [2]. Sorting the signed pilot frequencies gives a uniformly spaced sequence indexed by $m = 0, \ldots, 15$; the constant phase from the frequency origin is absorbed into each complex gain. The pilot-domain posterior mean is evaluated at these integer indices. Von Mises moments retain frequency uncertainty, and the noise update includes gain covariance and frequency-response uncertainty. Interpolation to noninteger indices requires an explicit physical delay branch, which lies outside the reported pilot-domain comparison.

The covariance term uses the paper's global component index; a source-compatible indexing option is available for inspection. Tests compare moments with independent numerical quadrature and support-update covariances with direct matrix inversions. The Python paper-equation implementation has not been verified for numerical equivalence with the original MATLAB code. Across the 2700 primary outputs, no run activated noise/prior variance-floor or Schur-denominator clipping. The implementation documentation records the high-concentration moment approximation and nonconcave frequency-update fallback. The differing support restrictions, observation budgets, posterior approximations, and stopping rules define the scope of the model comparison.

The 12 runs reaching the 500-iteration ceiling were replayed at the 5000-iteration ceiling used in the original VALSE simulation protocol [2]. Re-execution at 500 iterations first reproduced their saved outputs. Eleven capped runs then met the stopping criterion, bringing the total to 2699 of 2700; the remaining run was retained. The largest converged iteration count among these replays was 1140. The persistent capped case was a separated fractional profile at 10 dB (seed 1220261115, group index 19, zero-based). Using the original complete cell denominators, the largest absolute change in pilot NMSE was 0.000624 dB and that in mean noise ratio was 2.72×10^{-5}. These sensitivity results support the stability of the reported cell averages to the iteration ceiling. The main table uses the original 500-iteration setting throughout.

Appendix B. Continuous-delay optimization sensitivity

The bounded continuous-delay optimizer was checked at increased evaluation budgets. In a separate 600-call exploratory audit across all five profiles, 22 selected candidates reached the 60-evaluation limit; 20 of these occurred in dense profiles. Raising the budget to 300 resolved 18 of the 22 selected-candidate failures. This check assessed the control under a finite computation budget.

The independent CP experiment recorded diagnostics for all 10 800 refined fits. In 782 fits, at least one candidate reached the original evaluation limit; 121 selected a candidate with that status. Rerunning all 782 flagged fits with a 300-evaluation limit reduced selected-candidate failures to 18. Increasing the limit to 1000 for those 18 left four failures. Every estimate remained finite, and the two primary 20 dB CP-versus-residual gains changed by less than 10^{-4} dB. Re-execution reproduced the original noise arrays and flagged fits; unflagged per-group metrics were unchanged. The principal calibration contrast was stable under this budget check, with four residual stopping failures and possible local minima. The primary results retain the original predeclared 60-evaluation setting; the sensitivity analysis evaluates budget dependence rather than certifying a global optimum.

Data and code availability

All observations in this study are simulated. The accompanying local reproducibility package contains the generators, estimator implementations, random seeds, per-group metric arrays, aggregate numerical results, and figure scripts. The provenance of the original third-party VALSE demo is documented; the demo is not redistributed in the package.

CRediT authorship contribution statement

Yuxiao Kuang: Conceptualization, Methodology, Software, Formal analysis, Validation, Visualization, Writing – original draft, Writing – review & editing. Wenqing Liu: Validation, Formal analysis, Writing – review & editing. Wenjie Li: Validation, Visualization, Writing – review & editing.

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Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex was used for debugging assistance. The authors reviewed and verified the scientific content, code, results, references, and final text and take full responsibility for the content of the published article.

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